

NEW PRODUCTS

MAKE IN INDIA

COMPONENTS

Automotive microcontroller

The RL78/F24 and RL78/F23 microcontrollers (MCUs) designed for automotive actuator and sensor control applications support edge evolution in next-generation electronic and electrical architecture. The expanded RL78 family of low-power 16-bit



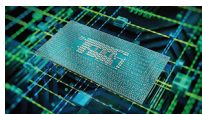
MCUs strengthens the company's automotive portfolio, offering customers highly reliable and high-performance solutions for systems ranging from actuators to zone control. Both MCUs support the CAN FD high-speed communication protocol and EVITA-Light security, delivering up to 70% faster operating frequencies, which can more than double the performance in brushless motor control applications.

Renesas Electronics

www.renesas.com

Mobile processors

The new 12th Gen Intel Core mobile processors are claimed to be the world's fastest mobile processors (40% faster than the previous generation), bringing their performance



hybrid architecture to mobile platforms. Intel has introduced 28 such mobile processors that deliver a feature-rich suite of computing capabilities, powering designs across consumer, enterprise, IoT, and other applications. Based on the Intel 7 process, new 12th Gen H-series processors offer up to 5GHz frequencies, 14 cores (6 P-cores and 8 E-cores), and 20 threads. They have nearly three times faster connectivity on exclusive

high-speed channels without legacy Wi-Fi interference.

Intel Corporation

www.intel.corporation

ADAS radar sensor

With the new AWR2944 radar sensor, Texas Instruments (TI) has expanded its extensive portfolio of analogue and embedded processing products and



technologies that can help automakers improve how advanced driver assistance systems

(ADAS) sense objects. The increased ability to quickly detect objects, monitor blind spots, and efficiently navigate turns and corners paves the way for automakers to achieve their vision of a collision-free future. The 77GHz AWR2944 radar sensor offers best-in-class radio frequency performance in a small form factor and provides 33% higher resolution than existing sensors. The sensor's unique hardware supported by Doppler division multiple access based signal processing improves the ability to sense oncoming vehicles at distances up to 40% farther away.

Texas Instruments

www.ti.com

Upgraded GaN ICs

The GaNFast power ICs with GaN-Sense technology have been upgraded



to increase efficiency, power density, and access additional fast-charger markets. GaNFast power ICs integrate GaN power and drive plus protection and control to deliver sim-

ple, small, fast, and efficient performance. The voltage rating of NV613x and NV615x GaNFast ICs has been upgraded from 650 to 700V for continuous operation and is rated at 800V for transient conditions. Increased voltage rating enables more efficient power transformer circuit designs and higher capabilities for areas of the world with unreliable, widely-varying power grids with extreme voltage spikes. Autonomous system-level monitoring and reaction ensures 'detect and protect' within 30ns—10x faster than for discrete implementations.

Navitas Semiconductor

www.navitassemi.com

Automotive positioning chip

The STA8135GA is an automotive-qualified GNSS (Global Navigation Satellite System) receiver that inte-



grates a triple-band positioning measurement engine on-chip, in addition to standard multi-band Position-Velocity-Time (PVT) and dead-reckoning. It helps driver-assistance systems make accurate decisions about the road ahead and delivers raw information for running any precise-positioning algorithm. The STA8135GA also enhances the performance of in-dash navigation systems, telematics equipment, smart antennas, and V2X communication systems, as well as marine navigation systems, drones, and other vehicles.

STMicroelectronics

www.st.com

5G smartphone chip

The Dimensity 9000 5G smartphone chip is built on the ultra-efficient TSMC N4 (4nm-class) production process for next-generation flagship smartphones, thus leading the